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(See 6.4)

MILITARY SPECIFICATION

POWER SUPPLY, METALLIC RECTIFIER GUIDED MISSILE FIRE CONTROL, NAVAL SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallic rectifier power supplies used to energize portions of guided missile fire control systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-S-3950 - Switches, Toggle, Environmentally Sealed, General Specification for.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

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MILITARY (Continued)

MIL-C-17588 - Circuit Breakers (Automatic - ALB-1) and Switch, Toggle (Circuit Breaker, Non-Automatic - NLB-1) Air, Insulated Housing, 125 Volts and Below, A.C. and D.C., (Naval Shipboard Use).
MIL-S-17773 - Switching Unit, Power Transfer (Bus Transfer) (Naval Shipboard Use).
MIL-M-28787 - Modules, Standard Electronic General Specification for.

STANDARDS

MILITARY

MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric).

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0967-LP-312-8010 - Electronic Equipment, Shipboard, Maintainability Design Criteria Handbook for Designers of.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The power supply shall be in accordance with class R of MIL-P-15736 and as specified herein.

3.2 Specific requirements. The power supply shall be as specified in table I.

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TABLE I. Specific requirements.

Parameter	Requirement
Input:	
A.c. V Frequency Phase	As specified in DOD-STD-1399, Section 300, type I power
Output:	
D.c voltage	Adjustable 26 to 34 V
D.c. current	40 A continuous 80 A peak, intermittent 2 second duration
Duty	Continuous
Enclosure	Dripproof protected
Type of load	Resistive, inductive, capacitive, and motor 0.9 power factor leading or lagging, as minimum
Mounting	Bulkhead
Meters	Voltmeter (0 to 50 V), ammeter (0 to 80 A, 3-1/2 inch round MIL-M-10304)
Ventilation	Natural draft
Ripple content, output voltage	1 percent, rms (maximum)
Ambient temperature range	23°C (room temperature) to 55°C

3.2.1 A.c. input. The power supply specified herein shall operate from type I power as described by DOD-STD-1399, Section 300.

3.2.1.1 Input power interruption. The power supply shall withstand input power interruption of duration ranging from 20 to 50 milliseconds (ms) resulting from transfer from one power source to an unsynchronized alternate power source. Equipment shall incur no damage to any component when the alternating current (a.c.) line input voltage to the equipment is interrupted over the range of time intervals specified above, and then restored. Upon reapplication of the 440 volt (V) a.c. input line, the direct current (d.c.) output voltage of the power supply is permitted to overshoot above the regulated set point of d.c. voltage, provided that output voltage does not exceed 35 V d.c. (see 3.2.7). Output voltage shall recover to the regulated set point within 200 ms following decay of transients. The blowing of fuses, solely as a result of the exercising of the automatic bus transfer switch, is prohibited (see 6.3).

3.2.1.2 Harmonic distortion of input current. With the exception of 11th and 13th harmonics of power frequency, the magnitude of any individual line input harmonic currents, from the second through the 32nd, expressed as a percentage of the full load fundamental root mean square (rms) line current, shall not exceed 3 percent for any input current between no load and rated full load value. Similarly, no individual harmonic line current from the 33rd harmonic through 20 kilohertz shall exceed a magnitude in percent equal to

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$1700/n^2$, where "n" is the harmonic multiple number. Harmonic currents shall be measured as specified in 4.2.7.

3.2.1.2.1 For any input current drawn from the 440-V line when the power supply is operated at output currents between no load and 40 amperes (A), harmonic line currents of order 11 and 13, expressed as a percentage of full load fundamental frequency rms line current, shall not exceed the percentages specified below:

(a) 6.5 percent for 11th harmonic.
(b) 4.5 percent for 13th harmonic.

3.2.1.2.2 The use of a.c. capacitors for filtering of input harmonic currents on the 440-V line is prohibited.

3.2.2 Regulation and adjustment.

3.2.2.1 Output voltage regulation. The d.c. output voltage measured at the power supply output terminals or load shall be automatically regulated within plus or minus 1 percent of the preset value selected from the voltage adjustability range specified in 3.2.2.2 despite variations in line input voltage, frequency, and ambient temperature, as permitted by 3.2, and changes in load current extending from no load to 40 A. When the power supply is operated at loads ranging from 40 to 45 A, the d.c. output voltage shall be regulated at the preset voltage within a plus 0.5 V, minus 1.5 V band despite the specified variations in line voltage, frequency, and ambient temperature.

3.2.2.2 Output voltage adjustment. Assuming a 4-V drop at full load, the d.c. output voltage measured at the load shall be continuously adjustable from 26 to 34 V d.c. at rated full load by means of an internally accessible rheostat which is mounted on the front panel of the enclosure. The voltage adjustment specified herein shall be accomplished with a resolution no greater than 0.25 V, and the voltage adjustment rheostat shall be equipped with a locking device to prevent changes in setting due to the effects of shock and vibration.

3.2.2.3 Remote sensing. The power supply shall regulate output d.c. voltage at a load point remote from the power supply. The line voltage drop from the power supply to the load may be any value ranging from 0 to 4 V at full load A. The power supply shall make provision for sensing of output load voltage by means of 2-wire remote voltage sensing leads which are hard-wired to a terminal block internal to the power supply enclosure. In the use of remote voltage sensing leads, the control of load voltage shall be such that the signal current in the sensing leads does not exceed 1 A under all specified operating conditions.

3.2.3 Output current. The rectifier equipment shall be rated at 40 A continuous duty.

3.2.4 Transient performance. D.c. output voltage measured at the terminals of the power supply shall not drop below 24.5 V upon step application of full load A to an equipment which initially operates to supply 28 V at no load. D.c. output voltage measured at the terminals of the power supply shall

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not rise above 35 V upon step removal of full load at the output terminals of an equipment which initially operates to supply 28 V d.c. at rated full load. The transient excursion in output terminal voltage caused by step application of load or step removal of load shall recover to, and remain within, the specified plus or minus 1 percent band of the initial voltage setting, within 200 ms following each above stated change in load.

3.2.5 Ripple voltage. The ripple content of the output voltage shall not exceed 0.5 percent of the preset d.c. voltage as an rms measurement. This ripple requirement shall apply to any load ranging from no load to 40 A resistive load, with balanced input a.c. line voltage.

3.2.6 Overcurrent operation. When operating into a load fault through line resistance which produces output voltage line drop of 4 V d.c. at 40 A (see 3.2.2.1), the power supply shall furnish sufficient output fault current to force the tripping of an ALB-1 circuit breaker conforming to MIL-C-17588 and having a trip element rated at 35 A. Overload faults, which load the output bus at 60 A, or 150 percent of rated full load current, shall be cleared without damage to any components of the power supply and without interruption of power supply operation by blowing of fuses or thermal overload trip of the input line controller. After the output load fault has been cleared by tripping of the circuit breaker, the d.c. power supply shall revert to voltage regulated operation in accordance with 3.2.2.1.

3.2.6.1 Overcurrent protection. The power supply shall be protected against overload (60 A or 150 percent load) and short circuit (step application of 128 A or 3.2 X rated full load) on output terminals. Protection shall be provided by fuses and input overload relays (see 3.2.10.2(a)) applied in accordance with MIL-P-15736. The use of electronic shutdown circuits which automatically disable the forward conduction of power semiconductors during output short circuit or overload is prohibited. Crowbar circuits are also prohibited as a method to protect the power supply against overload or short circuit on output terminals. The blowing of capacitor fuses is permitted for purposes of clearing a fault induced by step application of 128 A load (or 3.2 per unit load) on the output terminals of the power supply, which is initially at no load. Manual restart of the power supply is acceptable, in the event that input power has been interrupted by overload trip of the input line controller under conditions of overload on the output terminals of the power supply.

3.2.7 Overvoltage protection. Design of the power supply shall be such that failure of circuit components or connections between components, occurring singly or in combination, will not cause output voltage to exceed 35 V d.c. at the power supply.

3.2.8 Structureborne noise. Any structureborne noise generated by the equipment shall not exceed the noise acceptance levels of MIL-STD-740 for type 3 solidly mounted equipment under any specified operating conditions.

3.2.9 Construction.

3.2.9.1 Enclosure. The enclosure shall be dripproof protected with a hinged front access door. Meters shall be mounted on the door. Unless otherwise specified herein, switches and miscellaneous controls shall be mounted on the door for external operation.

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3.2.9.2 Mounting. The equipment shall be designed for bulkhead mounting. Bulkhead mounting dimensions shall be as shown on figure 1.

3.2.9.3 Access. The power supply shall pass through a 25-inch diameter hatch. In addition, the assembled power supply enclosure shall pass through a doorway 20 inches wide and 38 inches high with corners of 10-inch radius.

3.2.9.4 Size and weight. The maximum size and weight of the equipment shall be as follows:

Height - 24 inches
Width - 14 inches
Depth - 10 inches
Weight - 175 pounds

The above enclosure dimensions exclude extensions due to mounting supports.

3.2.10 Instruments and controls.

3.2.10.1 Meters. An ammeter and a voltmeter shall be provided for measuring the total output current and output voltage. Meters shall be ruggedized 3-1/2-inch round panel type in accordance with MIL-M-10304. The voltmeter shall have a graduated scale ranging from 0 to 50 V (full scale deflection). The ammeter shall have a graduated scale ranging from 0 to 80 A (full scale deflection) and shall operate from a 50-millivolt external shunt. Meters and shunts shall be furnished integral with the power supply.

3.2.10.2 Switches. The following switching facilities shall be provided:

(a) A 3-pole manual low voltage release effect (LVRE) controller shall be provided for on-off operation at the a.c. input. The controller shall provide thermal overload protection to the power supply, shall be flush mounted, and shall be in accordance with MIL-C-2212.
(b) An ammeter press-to-read switch shall be provided. The switch shall consist of a momentary contact toggle switch in accordance with MIL-S-3950. Normal position of switch contacts shall disconnect the ammeter from the load circuit and require operation of the switch to read the load current. The switch shall be a spring return to normal position.

3.2.10.3 Miscellaneous controls. The following miscellaneous controls shall be provided:

(a) An indicator light shall be provided and connected to indicate "power on" in the a.c. input circuit.
(b) A rheostat shall be provided for infinite adjustment of the output voltage over the specified range of 26 to 34 V. This rheostat shall be mounted on the interior surface of the hinged front panel and shall be accessible only when the door is open.

3.2.11 Reliability. Mean-Time-Between-Failure (MTBF) of the power supply shall be not less than 10,000 hours.

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3.2.12 Maintainability. The power supply shall be designed so that corrective maintenance can be performed by replacement of modular assemblies. The functional electronic circuitry of the power supply shall be implemented by means of Standard Electronic Modules having the design and physical dimensions specified in MIL-STD-1389. Modules shall be class I in accordance with MIL-M-28787 for submarine application. Modules introduced into the power supply shall be selected in accordance with specific detail requirements as defined by MIL-M-28787. Equipment design shall conform to the design-for-maintainability requirements of MIL-STD-470 and NAVSEA 0967-LP-312-8010.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-P-15736 and shall include the tests specified in 4.2.1 through 4.2.8.1 herein for selective, periodic and qualification inspection. The tests of 4.2.5.1 and 4.2.5.2 shall also be performed under routine inspection.

4.2 Tests.

4.2.1 Voltage regulation. The power supply shall be tested for steady state voltage regulation performance in accordance with the procedures of MIL-P-15736.

4.2.2 Transient regulation. For purposes of each transient regulation test specified herein, the power supply shall be operated with local sensing under each condition of line voltage, line frequency, and ambient temperature specified in MIL-P-15736 and 4.2.1 herein. The d.c. output voltage shall be adjusted at 28 V d.c. under each of the ten specified conditions of line voltage frequency, and ambient temperature. Transient voltage regulation tests shall be conducted as stated below in order to verify compliance with 3.2.4. During each of the specified tests, voltage deviation and transient recovery time shall be recorded by means of oscillograms of output voltage and load current having sufficient scale of amplitude and time to provide good readability:

(a) Transient regulation; application of load. The power supply shall initially be operated at no load. A resistive load of 40 A shall be abruptly applied as a step function load increase at the rectifier equipment output terminals. Application of the load shall be accomplished by operation of a magnetic contactor connected with the main contacts in series with the load.

(b) Transient regulation; removal of load. The power supply shall be operated under steady state conditions of voltage regulation into a 40 A resistive load applied at the output terminals of this equipment. This load shall suddenly be removed, so that a step function decrease in load from 40 A to no load is obtained. Interruption of the load shall be accomplished by operation of a magnetic contactor connected with the main contacts in series with the load.

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4.2.3 Overcurrent protection. The power supply shall be connected to output load terminals through the line resistances shown on figure 2. Load cable between power supply and load shall be sized at 35 A. Circuit breaker ALB-1, with trip element rated at 35 A d.c. shall be wired in series with resistor load bank, as shown on figure 2, and shall be closed. Switch S1 shall initially be open. Output voltage at the terminals of the power supply shall be 31.5 V d.c. The power supply shall be operated with local sensing to regulate d.c. voltage at the output terminals of the power supply equipment. The power supply shall supply 35 A to the load, which consists of load resistor R1 (0.8 ohm). Output voltage line drop between power supply and load shall be 3.5 V at 35 A d.c. Voltage developed across the load shall be 28 V d.c. when resistor R1 carries 35 A. When the test circuit of figure 2 has been established, switch S1 shall be closed. This switch closure, which places resistor R2 (0.8 ohm) in parallel with resistor R1 (0.8 ohm), shall cause the power supply to furnish 60 A overload current to the load, for purposes of tripping the ALB-1 circuit breaker. Fault current which flows through the circuit breaker shall be recorded by an oscillograph. The method by which the fault is cleared shall be recorded. The d.c. output voltage of the power supply shall be recorded after the fault has been cleared. The manner in which the fault is cleared and subsequent operation of the power supply shall demonstrate conformance to 3.2.6.

4.2.4 Overvoltage protection. Analysis of power supply design shall be performed in order to confirm that component failures, occurring singly or in combination, will not drive the power supply output voltage to a value greater than 35 V d.c. in compliance with 3.2.7.

4.2.5 Noise tests. Airborne and structureborne noise tests shall be performed on every power supply unit offered for delivery. In the course of both airborne and structureborne noise tests, the power supply shall be operated successively at no load and rated full load (40 A) in the voltage regulation mode conforming to 3.2.2.1.

4.2.5.1 Airborne noise tests. The test reports on the airborne noise characteristics of the power supply shall contain data in the form of tabulated readings and graphic level recordings on measured sound pressure levels and sound power levels. This data shall be accompanied by calculations describing the method of conversion of sound pressure data to sound power data.

4.2.5.2 Structureborne noise tests. The structureborne noise characteristics of the power supply shall be measured and reported in accordance with MIL-STD-740. These noise measurements and associated data shall demonstrate compliance with 3.2.8. During the performance of structureborne noise tests, the power supply equipment shall be solidly mounted at the points of attachment of the enclosure (see 3.2.9.2) on a test fixture which conforms to MIL-STD-740. The combined assembly of equipment and test fixture shall be resiliently mounted with the equipment oriented in its normal operating position for purposes of the structureborne noise test. Test data shall be submitted in the form of tabulated readings and graphic level recordings.

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4.2.6 Input power interruption. The power supply shall be energized through a power interrupt device consisting of a power transfer switch type ABT-A1 in accordance with MIL-S-17773 and rated at 100 A. The rectifier power supply shall be subjected to transfer and retransfer between unsynchronized a.c. voltage sources at each combination of output voltage setting and load specified in table II, Test Conditions No. 1 through 12 respectively. Ten cycles of transfer and retransfer of the d.c. power supply shall be performed at each test condition stated in table II. Each cycle is defined as follows:

(a) Transfer. The power supply shall initially be supplied by the normal a.c. voltage line. After steady state operation has been established, the power supply shall be transferred to the alternate a.c. voltage line.
(b) Retransfer. After 1 minute of steady state operation on the alternate a.c. voltage line, the power supply shall be retransferred from alternate to normal a.c. voltage line. The power supply shall continue to operate for 5 minutes on the normal a.c. voltage line before the cycle is repeated.

Output terminal voltage of the power supply shall be regulated by means of local sensing. Input voltage, power supply output terminal voltage, and output current shall be recorded by means of oscillograms having sufficient scale of amplitude and time to provide good readability. The recorded data shall demonstrate that the d.c. output voltage of the power supply equipment conforms to 3.2.2.1 during steady state operation following transfer and retransfer of the power supply between a.c. voltage lines.

TABLE II. Input power interruption.

Test condition no.	Period of power interruption	Load	Output voltage setting
1	Off-time for type ABT-A1 (100 A) is 20 -50 ms. Transfer time is 1/2 second for transfer and retransfer of d.c. power supply.	No load	32 V d.c.
2		No load	34 V d.c.
3		No load	28 V d.c.
4		No load	30 V d.c.
5		18 A d.c.	34 V d.c.
6		18 A d.c.	26 V d.c.
7		18 A d.c.	32 V d.c.
8		18 A d.c.	28 V d.c.
9		40 A d.c.	34 V d.c.
10		40 A d.c.	28 V d.c.
11		40 A d.c.	26 V d.c.
12		40 A d.c.	32 V d.c.

4.2.7 Harmonic distortion. Power supply equipment shall be tested to determine compliance with the limitations on allowable magnitudes of harmonic line current specified in 3.2.1.2.

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4.2.7.1 Input voltage distortion. The a.c. input line which feeds the power supply shall be monitored for distortion. For harmonics of order 2 through 46, no individual harmonic frequency component of a.c. line voltage shall exceed 3/4 percent of fundamental frequency line voltage, when the a.c. line supplies balanced full load line current to a 3-phase resistive load between 1 and 1.5 kilowatts.

4.2.7.2 Input current measurement. Output load current shall be varied from no load to rated full load in steps of 10 A d.c. At each load step commencing with no load, fundamental frequency component and harmonic components of input line current in lines A, B and C of the 440-V line shall be measured. Each harmonic line current of order 2 through 46 shall be recorded as rms A for each of lines A, B and C. Harmonic line current shall be plotted against line frequency. Any of the following methods of measurement of harmonic line current may be used:

(a) A noninductive line resistor may be inserted in each input line. Fundamental frequency and harmonic line currents shall be measured by determining the voltage developed across the line resistor. The resistance value of the line resistor shall be such that the potential drop at fundamental frequency will be 1 percent of line voltage.
(b) As an alternative to (a) above, current transformers may be used in order to monitor fundamental frequency and harmonic input line currents on the 440-V line.
(c) Fundamental frequency and harmonic input line currents shall be measured by means of current probe American Aerospace Model 938, or equal.

Accuracy of measurement of harmonic currents shall be 5 percent. A Hewlett Packard Model 3580 Spectrum Analyzer, or Spectral Dynamics Model SD 335A Spectrum Analyzer (or equal) shall be used for measurement of voltages resulting from sensing of line current.

4.2.8 Burn-in. The power supply shall be subjected to a burn-in test of 96 hours duration. The burn-in test shall be performed as a selective and qualification test in accordance with MIL-P-15736. Each power supply under test shall be operated for a period of 96 hours under the following conditions of voltage and load:

(a)	Output voltage and load:	28 V d.c. at 40 A d.c. initial loading at power supply terminals
(b)	Temperature:	Room ambient
(c)	Input line voltage:	440 V (nominal) with plus or minus 3 percent unbalanced between line-to-line voltages

The quality assurance tests, as specified in table III, shall be conducted during the 96 hour operational burn-in period at intervals as indicated:

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TABLE III. Burn-in tests.

Test	Test paragraph	Intervals
Input power interruption (test condition no. 11)	4.2.6	24 hours
Load cycling	4.2.8.1	12 hours
Random frequency vibration	See MIL-STD-781	

Random frequency vibration test shall consist of motion input of 6 g rms acceleration in accordance with MIL-STD-781. Duration of vibration shall be 10 minutes within each hour of equipment operating time. Output d.c. voltage at the output terminals of the power supply shall be monitored during the burn-in test in order to determine compliance with 3.2.2.1. Any component failures occurring during the burn-in of the power supply shall be reported, with explanation of cause of failure and recommendations for corrective action to prevent recurrence of the same failure.

4.2.8.1 Load cycling. The power supply shall be initially loaded at rated full load and shall be adjusted at 28 V d.c. regulated set point at the output terminals of the power supply. The load shall be cycled 24 times between rated full load and no load. Each cycle shall consist of operation at rated full load for an interval between 0.8 second and 1.2 seconds, followed by operation at no load for an interval of 10 seconds to complete the cycle. Voltage deviation and transient recovery time shall be recorded by means of oscillograms of output voltage and load current having sufficient scale of amplitude and time to provide good readability. The recorded data shall demonstrate that the power supply equipment conforms to 3.2.4.

5. PACKAGING

5.1 Preparation for delivery shall be in accordance with MIL-P-15736.

5.2 Use of polystyrene (loose-fill) material.

5.2.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise specified by the contracting activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When specified, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc., with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

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5.2.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler, and dunnage.

6. NOTES

6.1 The notes specified in MIL-P-15736 are applicable to this specification.

6.2 Ordering data. In addition to the ordering data requirements of MIL-P-15736, the acquisition documents should specify the following:

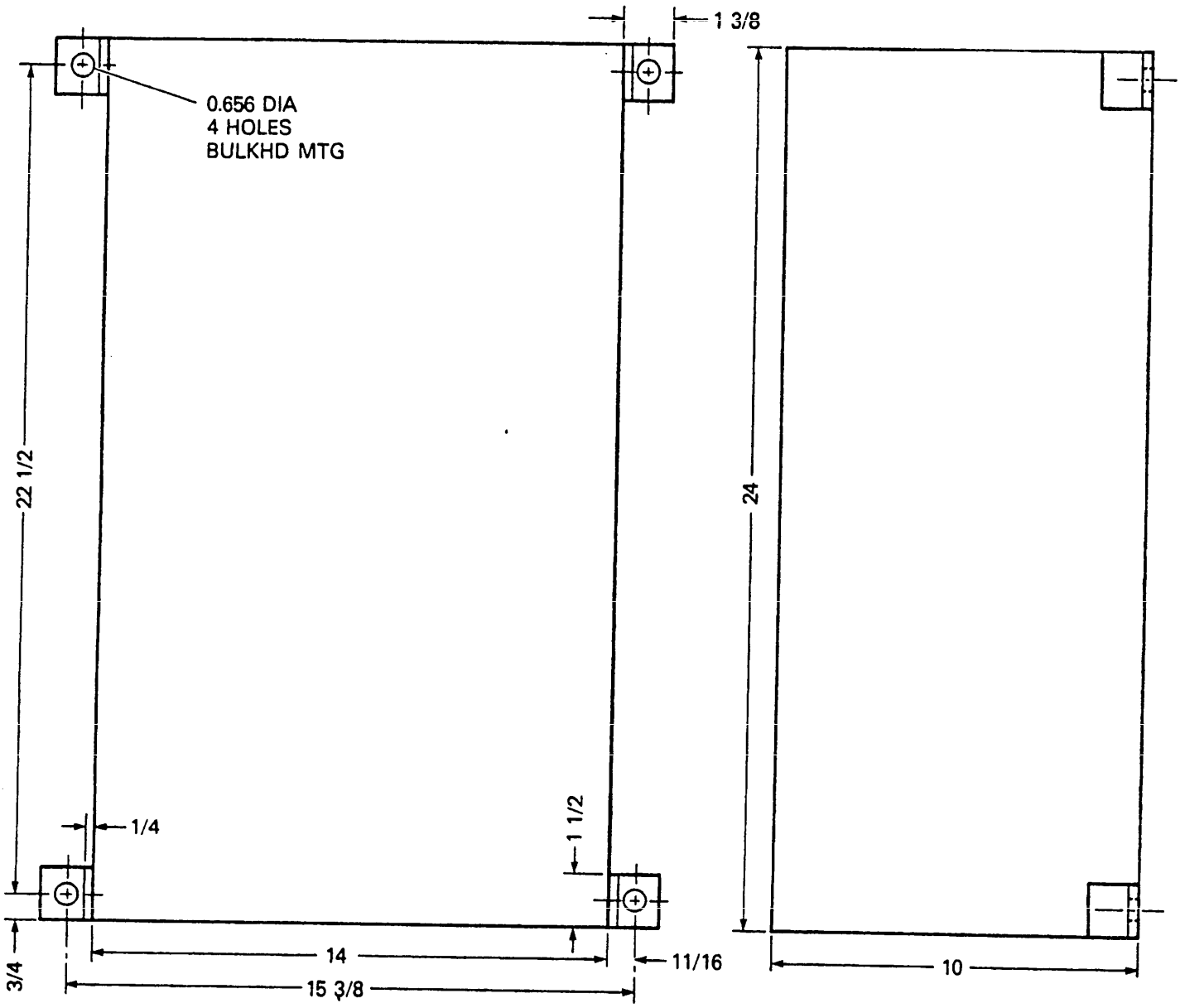
(a) When polystyrene "loose-fill" material is allowed (see 5.2.1).

6.3 Input power interruption. Missile fire control power supplies are a vital load on the submarine electric plant. In order to obtain maximum reliability and continuity of a.c. line voltage during battle or major casualties, all vital electrical loads are provided with limited break ship's service a.c. power. The missile fire control power supplies are supplied by two independent ship's service turbine generators (normal and alternate). Upon loss of the normal ship's service generator, vital loads are automatically transferred to the alternate ship's service generator (which is not synchronized with the normal ship's service generator) by means of automatic bus transfer (ABT) switches. After the normal ship's service generator has been restored, the ABT operates automatically to retransfer vital loads from alternate to normal ship's service generator. A time delay is introduced in the transfer operation of all bus transfer switches to avoid excessive current transients when switching from one generator to the other. A time interval between 100 and 250 ms is typically required to transfer the power supply from normal to alternate ship's service generator. (A similar time interval is required to retransfer the power supply from alternate to normal ship's service generator.) The fire control power supply experiences interruption of a.c. line voltage during ABT. The power supply is required to be designed so that when the 440 V, 3-phase, 60 hertz (Hz) line is interrupted and restored, no damage will result to any component (see 3.2.1.1). After restoration of a.c. line voltage, the fire control rectifier power supply is required to perform its intended function (see 3.2.1.1). Interruption of a.c. line voltage may occur during training exercises or electrical power system tests. Transfer of load from normal to alternate ship's service generator, and retransfer of load from alternate to normal ship's service generator, can be initiated manually, by operating the same bus transfer switch in manual mode with automatic function disabled.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6130-N230)

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NOTE: All dimensions are in inches.

SH 12108

FIGURE 1. Size and mounting dimensions of enclosure.

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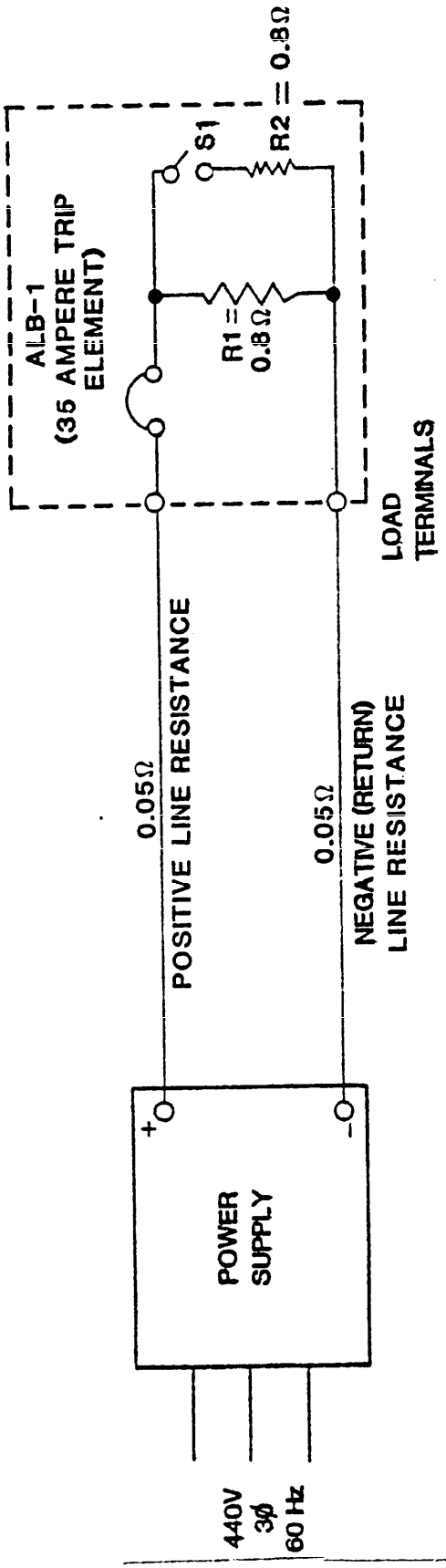


FIGURE 2. Test circuit for overcurrent protection.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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